

CAPSTONE PROJECT

AI-POWERED TECH CAREER RECOMMENDATION SYSTEM

Project Timeline: 2 Weeks

Project Deliverable: Submit a Jupyter Notebook (.ipynb) containing all code, visualizations, explanations, observations, and recommendations. Your notebook should be well documented and professionally organized.

PROJECT OVERVIEW

The aim of this project is to develop a Machine Learning-powered Career Recommendation System that predicts the most suitable technology career path for a user based on their interests, personality traits, technical preferences, work style, and learning preferences.

The system will use a Random Forest Classifier (Or any other algorithm of your choice, but state why the choice) to predict one of several technology career paths such as:

- **Data Science & Analytics**
- **Software Engineering**
- **Cybersecurity**
- **UI/UX Design**
- **Cloud Computing**
- **Product Management**

Students are encouraged to be creative and explore techniques that improve model performance and user experience.

1. DATA COLLECTION AND PREPARATION

I. Dataset Creation

Design and administer a questionnaire containing at least 15 questions covering:

- Demographic information
- Technical interests
- Personality traits
- Learning preferences
- Work style preferences
- Technology exposure

The final question should capture the user's preferred technology career path.

Target Variable: Career Path

Examples: Data Science & Analytics, Cybersecurity, Software Engineering, UI/UX Design, Cloud Computing, Product Management

II. Data Importation

- Import the collected dataset.
- Load the dataset into Pandas.

III. Initial Data Inspection

- Display the first five rows.
- Display the last five rows.
- Check dataset shape.
- Check column names.
- Review data types.
- Generate descriptive statistics.
- Identify categorical and numerical features.
- Check for duplicate records.
- Check for missing values.

IV. Data Cleaning

- Handle missing values.
- Remove duplicates.
- Correct inconsistent entries.
- Convert variables into appropriate formats.

2. EXPLORATORY DATA ANALYSIS (EDA)

Generate at least 5 meaningful insights supported by visualizations.

Suggested analyses:

- Distribution of career choices.
- Gender distribution.
- Age distribution.
- Relationship between coding interest and career choice.
- Relationship between design interest and career choice.
- Relationship between mathematical interest and career choice.
- Relationship between leadership preference and career choice.
- Correlation analysis.
- Feature importance exploration.
- Class distribution analysis.

Visualizations may include:

- Bar charts
- Pie charts
- Histograms
- Count plots
- Box plots
- Heatmaps

3. DATA PREPROCESSING AND FEATURE ENGINEERING

I. Handling Missing Values - Apply suitable methods:

- Mean
- Median
- Mode

- Forward Fill
 - Backward Fill
 - II. Encoding Categorical Variables - Convert categorical variables using:
 - Label Encoding
 - One-Hot Encoding
 - III. Feature Scaling - Experiment with:
 - StandardScaler
 - MinMaxScaler
 - IV. Feature Engineering - Students should attempt at least two techniques:
 - Feature Selection
 - PCA
 - Interaction Features
 - Aggregated Scores
 - Domain-specific feature creation
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4. MODEL DEVELOPMENT

- I. Train-Test Split - Split the dataset into Training Set and Testing Set. Suggested ratio: 80:20

- II. Model Selection

Primary Model: Random Forest Classifier

Students may compare performance against:

- Logistic Regression
- Decision Tree Classifier
- K-Nearest Neighbors
- XGBoost (Optional)

- III. Hyperparameter Tuning - Tune parameters such as:

- n_estimators
- max_depth
- min_samples_split
- min_samples_leaf
- max_features

Use: GridSearchCV, RandomizedSearchCV

5. MODEL EVALUATION

Evaluate the model using:

- Accuracy, Precision, Recall, F1-Score

Generate:

- Confusion Matrix
- Classification Report

For multiclass classification:

- Macro Average Metrics

- Weighted Average Metrics

Compare all models and select the best-performing model.

6. FEATURE IMPORTANCE ANALYSIS

Using Random Forest Feature Importance:

- Identify top 10 most important features.
 - Discuss why these features influence career recommendations.
 - Visualize feature importance using bar charts.
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7. RECOMMENDATION ENGINE

Develop a recommendation workflow:

Input: User responses to questionnaire.

Output:

- Recommended Career Path
- Prediction Confidence Score
- Suggested Learning Roadmap

Example:

Recommended Career: **Data Science & Analytics**

Confidence: **87%**

Recommended Learning Path: **Python, SQL, Statistics, Machine Learning, Power BI**

8. MODEL SAVING

Save the final trained model using Joblib or Pickle The model should be reusable for future predictions.

9. SUMMARY AND RECOMMENDATIONS

Summarize:

- Key EDA findings
- Most influential features
- Best-performing model
- Strengths of the system
- Limitations of the dataset

Provide recommendations for improving the project.

10. BONUS TASK (OPTIONAL)

Build a simple Streamlit Web Application. The application should:

- Accept user responses.
- Predict career path.
- Display confidence score.
- Display recommended learning roadmap.

Bonus marks will be awarded.